

E-commerce System Test Strategy Note (MFQ Framework)

1. Core Test Prioritization

The strategy is built on a risk-driven priority model, ensuring that the highest business risks are mitigated first. The priority order is:

1. Inventory Consistency (Highest Risk)
2. Order Accuracy & State Management
3. Payment Security & Fund Integrity
4. Secondary UI & Non-Critical Functions (Lowest Risk)

This prioritization is based on the potential impact of failure: overselling and inventory mismatches cause direct financial loss, customer complaints, and irreversible reputational damage, making them the non-negotiable "red lines" of the system.

2. Key Risks & Mitigation Focus

- **Overselling & Inventory Mismatch:** Addressed through strict testing of the inventory lock/deduct/restore flow, high-concurrency scenarios, and cross-system reconciliation between the storefront, order service, and WMS.
- **Fund Anomalies:** Covered by end-to-end payment/refund flow validation, idempotency testing, and three-way reconciliation of order, payment, and refund records.
- **State & Data Inconsistency:** Validated through order state machine testing, asynchronous message handling (duplicate/out-of-order messages), and eventual consistency verification.

3. Resource & Risk Trade-offs

Given limited test resources, the strategy adopts a clear trade-off model:

- **Prioritize Transaction Links:** 80% of test effort is allocated to the core transaction path (browse → add-to-cart → order → pay → ship → sign → refund), including functional, boundary, performance, and reliability testing.
- **De-prioritize Display Functions:** Non-critical UI/visual functions (e.g., product page styling, recommendation lists) are only covered by smoke tests in the main release cycle, with full regression deferred to future maintenance releases. This allows the team to focus on mitigating high-impact risks while meeting tight release timelines.